

$S_{\underline{a}}$  is closed subset of  $\mathbb{R}^n$ .

✓ claim:  $\{ \underline{a} = (a_1, a_2, \dots, a_n) \mid \underline{a}_j > 0 \ \forall j=1(1)n \}$ ,  $S_{\underline{a}}$  is bdd.  
for  $\Leftrightarrow S_{\underline{a}}$  ckt.

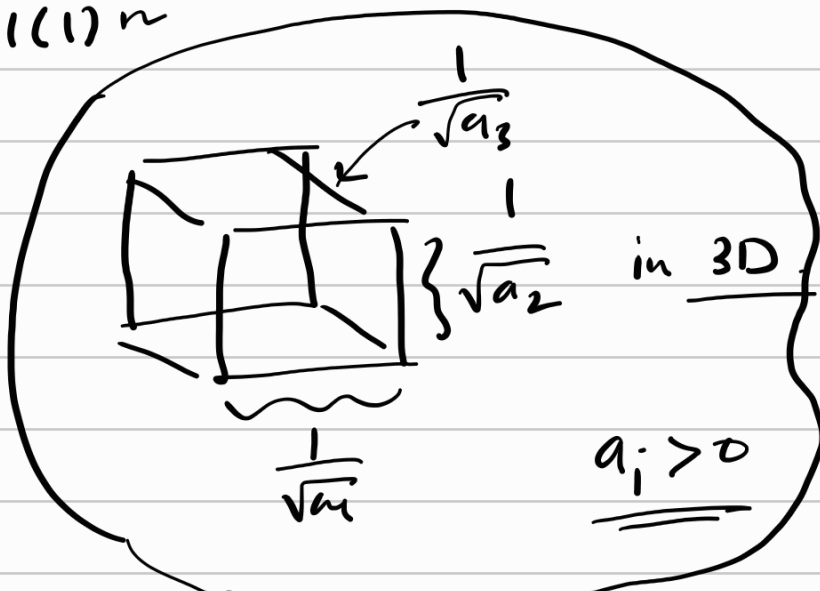
$$S_{\underline{a}} = \{ (x_1, x_2, \dots, x_n) \in \mathbb{R}^n : \sum_{j=1}^n a_j x_j^2 = 1 \}$$

$$x_j^2 = \frac{2}{a_j} \Rightarrow \sum_{\substack{k=1 \\ k \neq j}}^n a_k x_k^2 = -1 \quad (\Rightarrow \text{not possible})$$

$$a_j x_j^2 \leq 1 \quad \forall j=1(1)n$$

$$\Rightarrow x_j^2 \leq \frac{1}{a_j}$$

$$\Rightarrow |x_j| \leq \frac{1}{\sqrt{a_j}}$$



✓  $a_i = 0$ ,  $x_i \in \mathbb{R}$

$$\sum a_i x_i^2 = 1 \quad S_{\underline{a}} \text{ is not bdd}$$

✗  $\frac{a_i < 0}{a_i > 0}$

$x_i \in \mathbb{R}$   
 $x_j \in \mathbb{R}$

$$F = \{ \underline{a} \in \mathbb{R}^n \mid a_i > 0 \}$$

